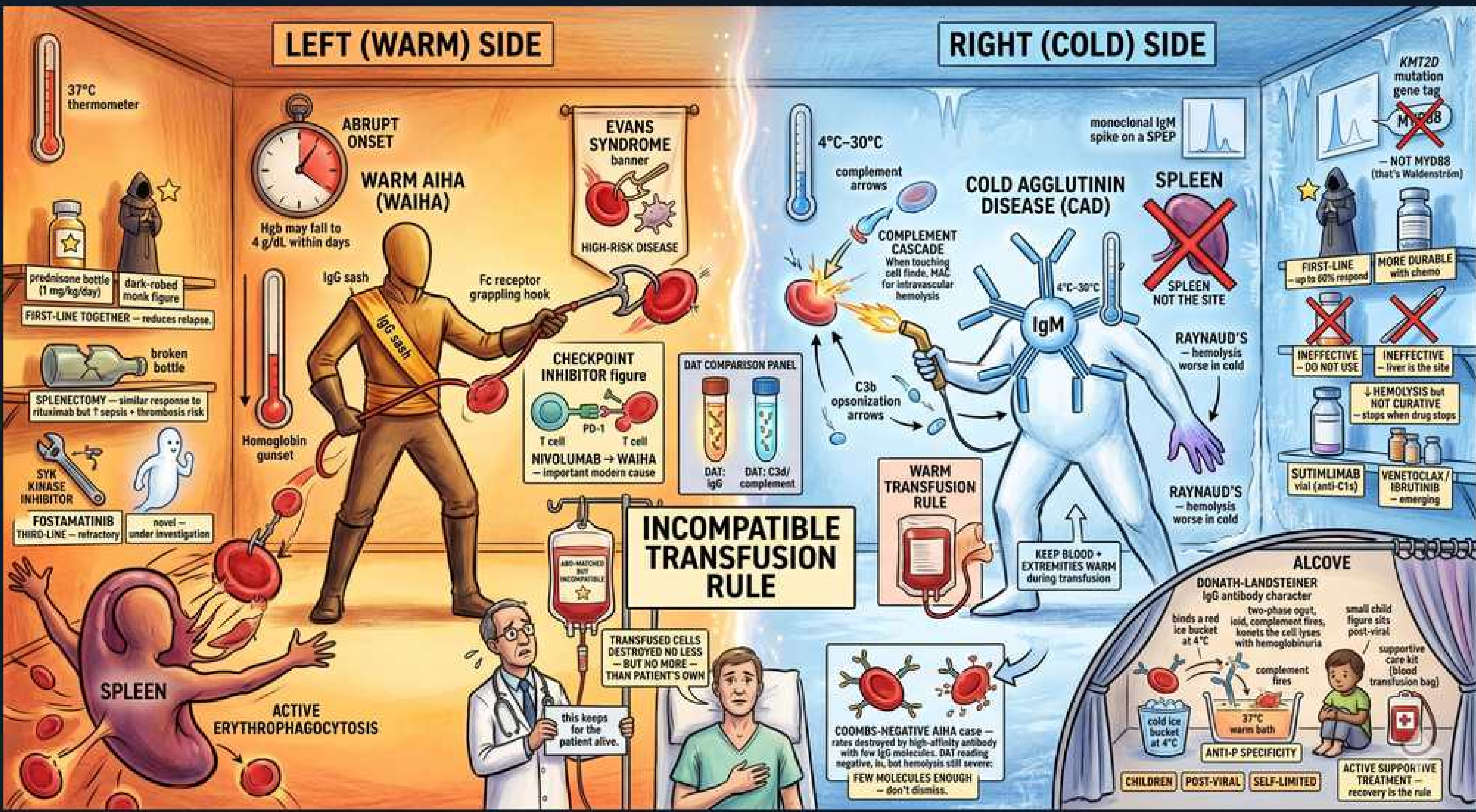
Autoimmune Hemolytic Anemia — Warm (IgG) vs Cold (IgM)



1. Warm AIHA (WAIHA) — IgG, 37C

WHAT YOU SEE

Left warrior labeled WAIHA

WHAT IT ENCODES

Warm AIHA is mediated by IgG that binds RBCs maximally at body temperature (37C). It is the most common AIHA. DAT is positive for IgG (+/- C3). Hemolysis is EXTRAVASCULAR in the spleen.

BOARD TRAP

Warm AIHA = IgG, extravascular, splenic — contrast cold (IgM, intravascular/hepatic complement). The thermometer (37C) is the key.

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2. IgG / Fc receptor grappling — extravascular

WHAT YOU SEE

IgG-coated RBC grabbed by Fc hook

WHAT IT ENCODES

IgG-coated RBCs are partially phagocytosed by splenic macrophage Fc receptors, producing SPHEROCYTES and extravascular hemolysis. The spleen is the site of destruction.

BOARD TRAP

Spherocytes appear in BOTH warm AIHA and hereditary spherocytosis — the DAT (positive in AIHA, negative in HS) distinguishes them.

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3. Active erythrophagocytosis — spleen

WHAT YOU SEE

Spleen consuming RBCs

WHAT IT ENCODES

In warm AIHA the spleen is the main graveyard, so SPLENECTOMY is an effective second-line therapy after steroids. Rituximab is an alternative steroid-sparing option.

BOARD TRAP

For warm AIHA the spleen IS the site (splenectomy helps); for cold agglutinin disease the spleen is NOT the site (splenectomy useless).

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4. Steroids — first-line warm AIHA

WHAT YOU SEE

Prednisone bottle, left side

WHAT IT ENCODES

Glucocorticoids are FIRST-LINE for warm AIHA, reducing autoantibody production and macrophage Fc function. Rituximab or splenectomy follow if steroids fail or can't be tapered.

BOARD TRAP

Steroids work for WARM AIHA but are generally INEFFECTIVE for cold agglutinin disease — a key treatment discriminator.

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5. Evans syndrome

WHAT YOU SEE

Banner: Evans syndrome

WHAT IT ENCODES

Evans syndrome is warm AIHA combined with immune thrombocytopenia (+/- immune neutropenia), often associated with autoimmune disease, lymphoma, or immunodeficiency.

BOARD TRAP

Coombs-positive hemolysis PLUS isolated low platelets = Evans; look for an underlying lymphoproliferative or autoimmune disorder.

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6. Checkpoint inhibitor / drug triggers

WHAT YOU SEE

Drug-trigger panel (nivolumab etc.)

WHAT IT ENCODES

Warm AIHA can be secondary to drugs (checkpoint inhibitors, methyldopa), CLL/lymphoma, SLE, or infection. Always evaluate for a secondary cause and the offending drug.

BOARD TRAP

New AIHA on a checkpoint inhibitor is drug-induced immune hemolysis — stop the drug; it isn't always idiopathic.

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7. Cold agglutinin disease (CAD) — IgM

WHAT YOU SEE

Right blue figure labeled CAD, IgM

WHAT IT ENCODES

Cold agglutinin disease is IgM-mediated, active at 4-30C. IgM fixes complement; DAT is positive for C3d (not IgG). Causes RBC agglutination and acrocyanitic symptoms in the cold.

BOARD TRAP

CAD DAT is C3-positive, IgG-negative — opposite the IgG-positive pattern of warm AIHA.

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8. Complement cascade — C3b opsonization

WHAT YOU SEE

Complement arrows on RBC

WHAT IT ENCODES

IgM activates complement; RBCs coated with C3b are cleared by the LIVER (Kupffer cells) — extravascular hepatic hemolysis — with some intravascular lysis when full complement assembles.

BOARD TRAP

Cold AIHA destruction is hepatic/complement-mediated, NOT splenic — which is why splenectomy doesn't help CAD.

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9. Cold 4-30C thermometer

WHAT YOU SEE

Cold thermometer

WHAT IT ENCODES

Cold agglutinins bind RBCs at low temperatures, so symptoms (acrocyanosis, hemolysis) worsen with cold exposure. Keeping the patient WARM is foundational management.

BOARD TRAP

First-line cold AIHA management is avoidance of cold/keeping warm — not steroids, which work poorly here.

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10. Raynaud / acrocyanosis

WHAT YOU SEE

Blue fingers, hemolysis worse in cold

WHAT IT ENCODES

Cold-induced agglutination causes Raynaud-like acrocyanosis and hemolysis on cold exposure; classically follows Mycoplasma pneumoniae (anti-I) or EBV (anti-i) infection, or underlying lymphoma.

BOARD TRAP

Anti-I = Mycoplasma; anti-i = EBV/infectious mononucleosis — a tested antigen-specificity pairing.

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11. Keep blood warm during transfusion

WHAT YOU SEE

Warm transfusion rule banner

WHAT IT ENCODES

In CAD, transfused blood must be WARMED to avoid triggering agglutination/hemolysis; complement-directed therapy (sutimlimab, anti-C1s) is now available for chronic CAD.

BOARD TRAP

Cold AIHA needs blood warming and complement-targeted therapy; standard steroids/splenectomy used in warm AIHA generally fail.

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12. Sutimlimab — anti-C1s

WHAT YOU SEE

Sutimlimab vial

WHAT IT ENCODES

Sutimlimab blocks classical-pathway complement at C1s, preventing C3b opsonization and hemolysis in cold agglutinin disease — a targeted therapy where steroids are ineffective.

BOARD TRAP

Sutimlimab is for COLD agglutinin disease (complement-mediated); it has no role in IgG-driven warm AIHA.

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13. Incompatible transfusion rule

WHAT YOU SEE

Center banner: incompatible transfusion

WHAT IT ENCODES

In severe AIHA, autoantibody can mask alloantibodies and make crossmatch difficult, but NEVER withhold life-saving transfusion — give the best-matched units for the patient's anemia.

BOARD TRAP

Don't deny urgent transfusion in AIHA over crossmatch incompatibility; transfuse the least-incompatible blood when life-threatened.

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14. Donath-Landsteiner — paroxysmal cold hemoglobinuria

WHAT YOU SEE

Alcove: anti-P antibody, children

WHAT IT ENCODES

Paroxysmal cold hemoglobinuria is caused by a biphasic Donath-Landsteiner IgG anti-P antibody that binds in the cold and lyses on warming; classically post-viral in CHILDREN, usually self-limited.

BOARD TRAP

PCH is IgG (Donath-Landsteiner, anti-P), NOT IgM — a cold hemolysis that is antibody-class opposite of classic cold agglutinin disease.
